Koushik Padmanabhan

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EDUCATION

University of Southern California Aug 2026 – May 2028
M.S. Computer Science (Game Development) *Los Angeles, CA*

Sri Ramachandra Institute of Higher Education and Research Aug 2021 – May 2025
B.Tech, Computer Science (AI & ML) — CGPA: 9.09/10 (First Class with Distinction) *Chennai, India*

TECHNICAL SKILLS

Languages: Python, C#, C++, JavaScript, Perl, Bash, SQL
Game & Simulation: Unity (C#), Unreal Engine (Blueprints), Godot, Unity ML-Agents
AI/ML: PyTorch, TensorFlow, Stable-Baselines3, OpenCV, scikit-learn, HuggingFace Transformers
Systems & Tools: FastAPI, Flask, React, Node.js, Docker, Git, MongoDB, MySQL, MSSQL, n8n

GAME AI & RESEARCH PROJECTS

Reinforcement Learning Agents
huggingface.co/koushik2k3

- Trained and published 7 RL agents to the Hugging Face Hub with reproducible model cards and evaluation scores, spanning value-based methods (Q-learning on FrozenLake and Taxi, DQN on Atari Space Invaders) and policy gradients (PPO on LunarLander).
- Built Unity ML-Agents policies across three control regimes: continuous locomotion and object fetching (Huggy), aim-and-fire targeting under a moving objective (SnowballTarget), and sparse-reward two-stage navigation (Pyramids), where curiosity-driven intrinsic rewards were needed to solve the button-then-goal sequence.
- Tuned reward shaping, observation stacking, and PPO hyperparameters per environment; agents are playable in-browser through the Unity ML-Agents demo spaces.

Unreal Traffic Simulator | Unreal Engine, Blueprints

- Built a multi-agent traffic simulation in Unreal with Blueprint-driven vehicle AI: agents follow authored two-lane spline paths and brake reactively when a vehicle enters their forward detection range, resolving conflicts without scripted waypoint timing.
- Implemented a lane re-acquisition behavior so vehicles spawned or displaced off-road locate the nearest valid lane and merge back into traffic flow, keeping the simulation stable under arbitrary spawn placement.

Unity & Godot Game Suite | Unity (C#), Godot, itch.io

- Shipped 8 playable prototypes across Unity and Godot, spanning an endless aircraft dodger with procedural obstacle spawning and parallax scrolling, a wave-based physics arena where enemy count doubles per wave, a competing-objective football game, a Fruit Ninja–style slicer, a side-scrolling runner, twin-stick shooters, and a 2D platformer with collectibles and enemy hazards.
- Implemented core gameplay systems throughout: input handling, spawn and difficulty pacing, physics-based push and collision resolution, and win/loss state management.

FocusNet: Deep Learning Framework for Fundus Pathology
https://link.springer.com/chapter/10.1007/978-3-032-06253-6_36

- Designed FocusNet, a DenseNet169 backbone with a 1x1 pixel-wise attention module, classifying fundus images into normal, cataract, glaucoma, and diabetic retinopathy at 95% accuracy on the 4,000-image Kaggle Eye Diseases Classification dataset.
- Benchmarked 12 configurations across ResNet-50/101, MobileNetV1/V2, and DenseNet-121/169; attention lifted DenseNet169 from 90% to 95% accuracy and diabetic-retinopathy F1 from 0.79 to 0.92. Published in Computer Vision and Robotics (CVR 2025), Springer, pp. 455-467.

EXPERIENCE

Mindera Feb 2025 – Jun 2026
Software Engineer *Bangalore, India*

- **Built** XGBoost-based anomaly detection pipelines for a fraud detection service, screening 1,000 transactions/day at 97% precision.
- **Spearheaded** an internal onboarding assistant that lets new developers query the codebase in natural language and get answers with source-document citations, then scaffold their first change through a built-in feature command; built on Spec Kit, it cut ramp-up from knowledge transfer to first contribution by 50%.
- **Rebuilt** the new-developer Docker setup after a migration left artifacts missing, replacing manual database dump downloads with automated setup SQL scripts and reducing environment setup from 2 hours to 30 minutes.

Katpro Technologies Dec 2023 – Dec 2024
Software Developer (Part-time) *Remote*

- Built a full-stack clinical coding application (React, FastAPI, Ollama) that extracts physician notes via OCR and uses a locally-hosted LLM with retrieval-augmented generation to return ICD-10 and CPT billing codes.
- Architected an end-to-end leasing automation platform for an apartment-finding client, chaining AI voice, email, and SMS agents to take renters from inbound call to signed lease with no human coordination.
- Created an OCR + LLM document pipeline extracting purchase orders, invoices, and GRNs at 95% field-level accuracy; deployed on Python, React, and MSSQL, replacing 20 hours of manual entry per week.
- Automated resume screening with LLM-based scoring and role-based routing via n8n, cutting time-to-shortlist by 60%.